Huawei Access Node BAS-ACC-05 Basavanagudi POP Hardware/Environmental Fault Resolution

Node: BAS-ACC-05 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_5134

Description

The occurrence of ASIC memory parity errors, line card crashes, or power supply redundancy loss on Huawei Access Node BAS-ACC-05 at Basavanagudi POP is primarily attributed to the intricate interplay between hardware and environmental factors. Specifically, the failure domain is rooted in the Line Card's (LC) ability to manage massive amounts of data transmitted over the network, which can lead to memory parity errors. Furthermore, the Power Supply Redundancy (PSR) mechanism, designed to ensure continuous operation in the event of a power supply failure, can be compromised by various environmental factors such as voltage fluctuations, overheating, or humidity. In the context of the Huawei Access Node, the Line Card's ability to detect and respond to these faults is critical, and any failure to do so can result in the propagation of alarms, such as ALARM_5134, indicating a severity level of Warning. It is essential to note that the root cause of these failures can be multifaceted, involving both hardware and environmental factors, and thus, a comprehensive understanding of the system architecture, as well as the interplay between various components, is crucial for effective resolution.

Pre-requisite Checks

- Verify that the remote hands have been dispatched to Basavanagudi POP.

- Confirm that the redundant link status is operational and not experiencing any issues.

- Ensure that the power supply redundancy is functioning correctly and that there are no signs of overheating or voltage fluctuations.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to high temperatures	High
Line card crashes caused by software bugs	Medium
Power supply redundancy loss due to environmental factors	Low

Diagnostic Commands & Expected Outputs

```
Command: display power supply, Expected Output: Power supply status: normal, Description: This command is used to verify the status of the power supply redundancy, ensuring that it is functioning correctly and that there are no signs of overheating or voltage fluctuations.

Command: display line-card, Expected Output: Line card status: normal, Description: This command is used to verify the status of the Line
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Card, ensuring that it is functioning correctly and that there are no signs of memory parity errors or crashes.

Command: display system internal alarm, Expected Output: ALARM_5134: Warning, Description: This command is used to verify the presence of the ALARM_5134, indicating a severity level of Warning, and to determine the root cause of the failure.
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Resolution Steps

Step 1: Step: Replace faulty Line Card, Command: replace line-card 1/1/1, Description: This step involves replacing the faulty Line Card with a new one, ensuring that the system is restored to a normal operating state.

Step 2: Step: Reboot the Access Node, Command: reboot system, Description: This step involves rebooting the Access Node to ensure that the system is restored to a normal operating state and that any remaining issues are resolved.

Step 3: Step: Verify the Power Supply Redundancy, Command: display power supply, Description: This step involves verifying the status of the Power Supply Redundancy, ensuring that it is functioning correctly and that there are no signs of overheating or voltage fluctuations.

Post-Check Verification

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Command: display system internal alarm, Expected Output: No alarms present, Description: This command is used to verify that the ALARM_5134 has been cleared and that the system is operating normally.

Command: display power supply, Expected Output: Power supply status: normal, Description: This command is used to verify the status of the power supply redundancy, ensuring that it is functioning correctly and that there are no signs of overheating or voltage fluctuations.
```

Roll-back Procedures

- Step: Restore the original Line Card, Command: replace line-card 1/1/1, Description: This step involves restoring the original Line Card, ensuring that the system is returned to its previous state.

- Step: Reboot the Access Node, Command: reboot system, Description: This step involves rebooting the Access Node to ensure that the system is returned to its previous state.